

Scientific Computation

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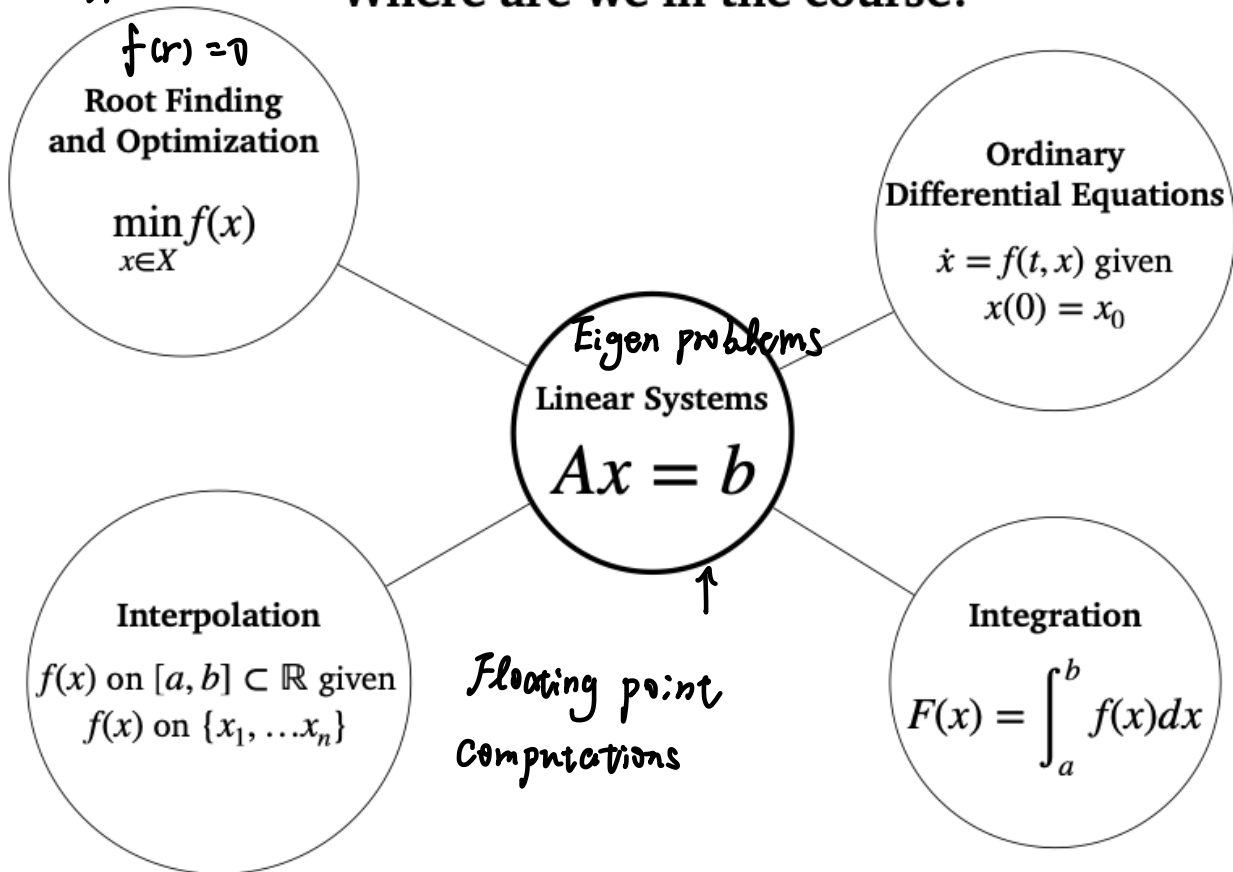
Lecture 20 - SVD application & Newton's Method

Lecture Outline

- The Singular Value Decomposition (SVD)
- Visualizing the SVD

- Applications of the SVD
- Newton's method

non linear Where are we in the course?



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Root Finding $f(r) = a \rightarrow f(r) - a = 0$

A common problem in scientific computation is to find a **root** (i.e., a zero) r of some function f :

$$f(r) = 0.$$

→ scalar case ($f: \mathbb{R} \rightarrow \mathbb{R}$)

↳ f may have no zeros, one zero or many

↳ the linear/affine case is now easy to us.
we have spend most of this covering so far
($Ax=b$, $f(x)=Ax-b=0$)

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Newton's Method

We will derive an iterative scheme from the first order Taylor series approximation about an iterate x_k :

$$0 = f(r) = f(x_k) + f'(x_k)(r - x_k) + O(|r - x_k|^2).$$

$E_2 = \downarrow$

$$\approx f(x_k) + f'(x_k)(r - x_k)$$

$$\hookrightarrow \frac{f''(\xi)(r - x_k)^2}{2}$$

Define x_{k+1} as the solution r to this linearized

equation:

$$f(x_k) + f'(x_k)(x_{k+1} - x_k) = 0$$

$$\Rightarrow \underline{x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}}$$

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Geometric Interpretation of Newton's Method

Given some initial iterate x_0 , we compute

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}.$$

The tangent line at $(x_0, f(x_0))$ is $l(x) = f(x_0) + f'(x_0)(x - x_0)$
(our Taylor series 1st order approximation) We find x , such
that the tangent $l(x) = 0$ (cross the x -axis). Repeat for
 $x_2, x_3, \dots$

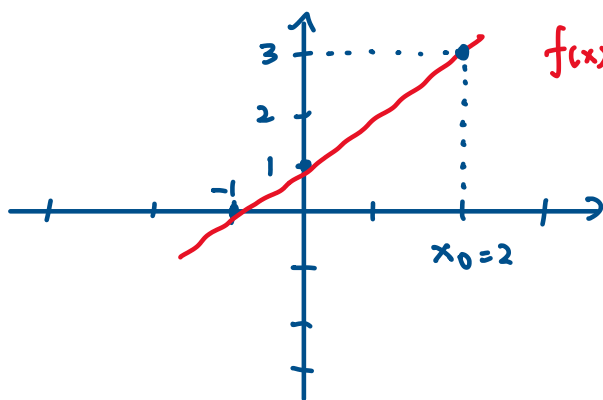
Questions: $\hookrightarrow$ How do we choose x_0 ?

$\hookrightarrow$ Will this converge to a root $f(x_k) = 0$

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Exactness for an Affine Function

Example 1: Consider the function $f(x) = x + 1$. Apply Newton's method to solve it
for $x_0 = 2$.



$$\begin{aligned} x_0 &= 2 \\ f'(x_0) &= 1 \\ x_1 &= x_0 - \frac{f(x_0)}{f'(x_0)} \\ &= 2 - \frac{3}{1} \\ &= -1 \end{aligned}$$

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A Nonlinear Example $f(r) = 0$

Example 2: Consider $f(x) = x^2 - a$ for $a > 0$. Write out the procedure for the k th iteration of Newton's method and run three iterations of the algorithm for $a = 2$ and $x_0 = 3$.

$$f(x) = x^2 - 2 = 0 \Rightarrow \text{the solution is } x \pm \sqrt{2}$$

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)} = x_k - \frac{x_k^2 - a}{2x_k}$$

$$= x_k - \frac{x_k}{2} + \frac{a}{2x_k} = \frac{1}{2} \left(x_k + \frac{a}{x_k} \right), \quad a = 2$$

$$x_1 = \frac{1}{2} \left(3 + \frac{2}{3} \right) = \frac{11}{6}$$

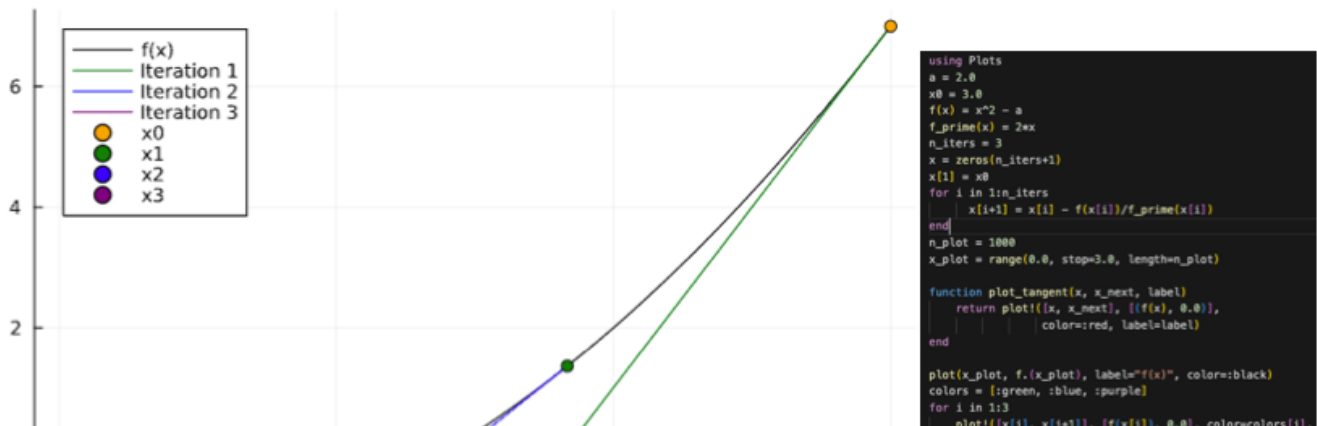
$$x_2 = \frac{1}{2} \left(\frac{11}{6} + \frac{2}{\frac{11}{6}} \right) \approx 1.46212 \dots$$

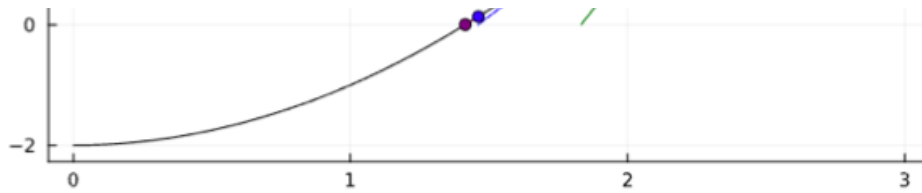
$$x_3 = \frac{1}{2} \left(x_2 + \frac{2}{x_2} \right) \approx 1.415$$

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Visualizing Newton's Method

Let's watch this algorithm in action and try to understand it geometrically:





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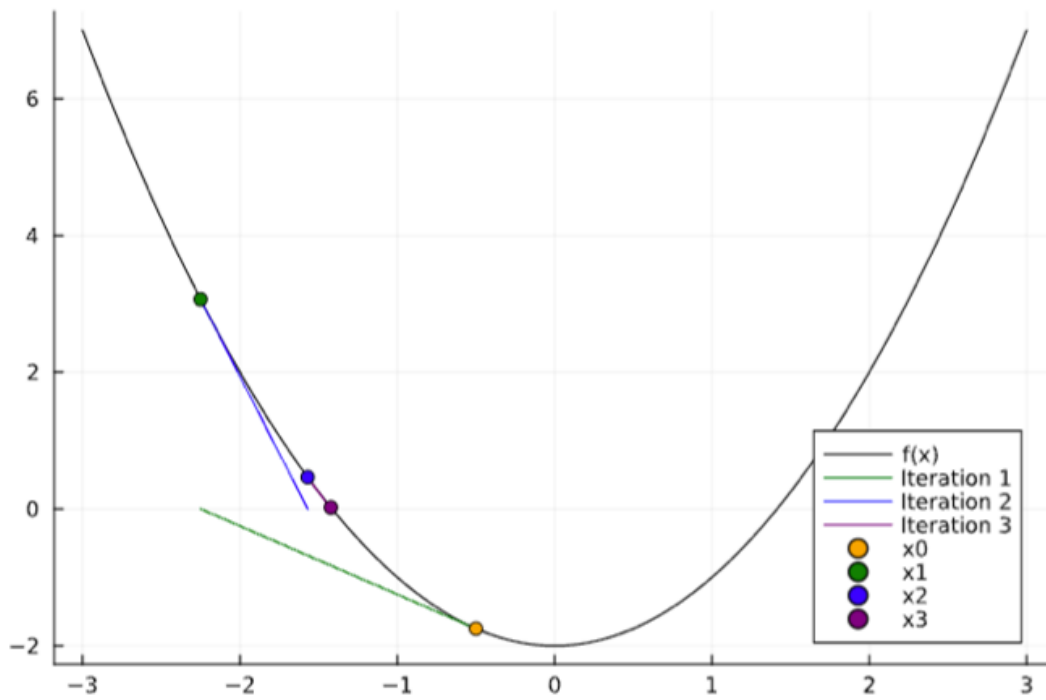
label="Iteration $(i)$"
end
scatter([x0], [f(x0)], label="x0", color=orange)
scatter([x2], [f(x2)], label="x2", color=green)
scatter([x3], [f(x3)], label="x3", color=blue)
scatter([x4], [f(x4)], label="x4", color=purple)

```

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Another Initial Point

If $x_0 = -0.5$, we get an entirely different set of iterations that finds another root!



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Division Without Division

Example 3: Let's write an algorithm for determining a/b without division.

$\frac{a}{b} = a \cdot \frac{1}{b}$, to find $\frac{1}{b} = x$ (and then multiply $a \cdot x$)

$f(x) = \frac{1}{x} - b = 0$, now solve $f(x) = 0$ with Newton's method:

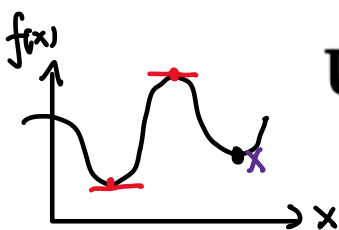
$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)} = x_k - \frac{\frac{1}{x_k} - b}{-\frac{1}{x_k^2}} = x_k + x_k^2(b - \frac{1}{x_k})$$

$$\begin{aligned}
 x_{k+1} - x_k &= \frac{f'(x_k)}{\left(-\frac{1}{x_k^2}\right)} \\
 &= x_k + x_k - b x_k^2 \\
 &= 2x_k - b x_k^2 \\
 &= \underbrace{x_k(2 - b x_k)}_{\text{involves no division}}
 \end{aligned}$$

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Lecture Outline

- Visualizing Newton
- Optimization with Newton
- Convergence of Newton's method



Unconstrained Scalar Optimization

objective / const. $\max f(x) \Leftrightarrow \min -f(x)$

We often want to find the minimum of a function f :

$$\min_{x \in \mathbb{R}} f(x)$$

We know that the first order optimality condition is $f'(x) = 0$ for critical points (which includes minima and maxima).

Let's apply Newton's method to find roots of $f'(x)$

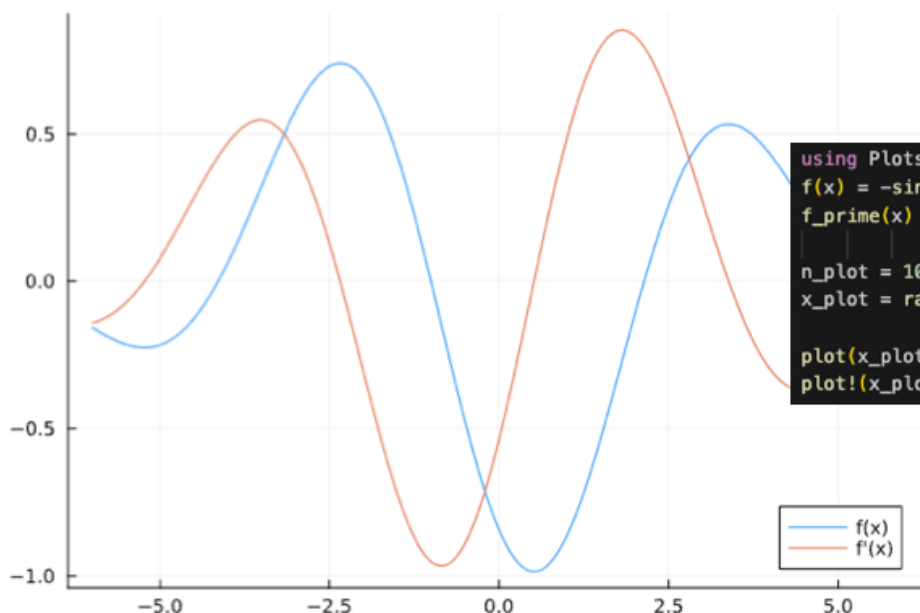
$$x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)}$$

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Optimization Example

Example 4: Consider $f(x) = -\sin(x+1)e^{-\frac{x^2}{20}}$ with derivative

$$f'(x) = e^{-\frac{x^2}{20}}\left(\sin(x+1)\frac{x}{10} - \cos(x+1)\right).$$



```
using Plots
f(x) = -sin(x + 1.0)*exp(-x^2/20.0)
f_prime(x) = exp(-x^2/20.0)*(sin(x+1)*(x/10) - cos(x+1))
n_plot = 1000
x_plot = range(-6.0, stop=6.0, length=n_plot)

plot(x_plot, f.(x_plot), label="f(x)")
plot!(x_plot, f_prime.(x_plot), label="f'(x)")
```

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Another Way of Thinking About Newton's Method for Optimization

Let's get the same result from another angle. Start by writing out the second order Taylor series approximation of $f(x)$ about x_0 :

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2} f''(x_0) (x - x_0)^2 + O(|x - x_0|^3)$$

$$\Rightarrow f(x) \approx m_0(x) \triangleq f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2} f''(x_0)(x - x_0)^2$$

now find a critical point of our quadratic model

$m_0(x)$:

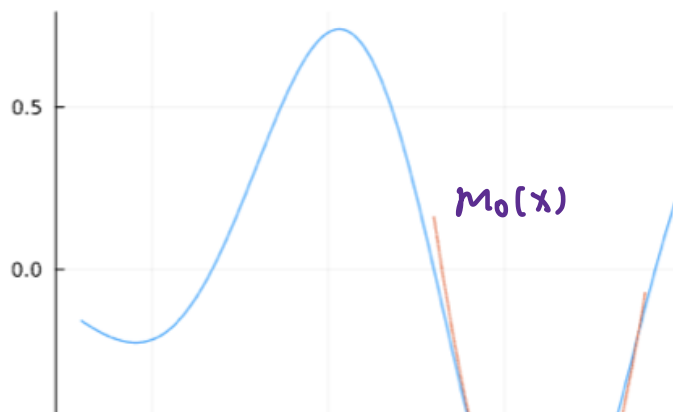
$$0 = m_0'(x) = f'(x_0) + f''(x_0)(x - x_0)$$

$$x - x_0 = \frac{-f'(x_0)}{f''(x_0)} \Rightarrow x_1 \triangleq x_0 - \frac{f'(x_0)}{f''(x_0)}$$

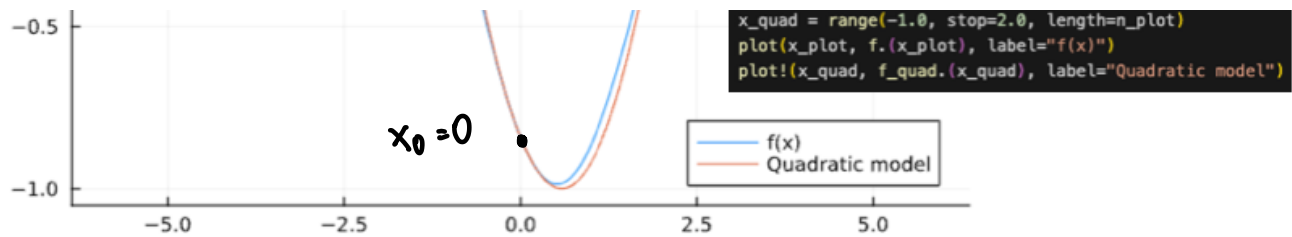
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Quadratic Model Visualized

Here's what a quadratic model of $f(x)$ at $x_0 = 0.0$ looks like:



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using Plots
f(x) = -sin(x + 1.0)*exp(-x^2/20.0)
f_prime(x) = exp(-x^2/20.0)*(sin(x+1)*(x/10) - cos(x+1))
f_prime_prime(x) = exp(-x^2/20.0)*(sin(x+1)/10 + cos(x+1)*x/10 - sin(x+1)*x^2/100 + sin(x+1) + cos(x+1)*x/10)
x0 = 0.0
f_quad(x) = f.(x0) + f_prime.(x0).*(x.-x0) .+ 0.5*f_prime_prime.(x0).*(x.-x0).^2
n_plot = 1000
x_plot = range(-6.0, stop=6.0, length=n_plot)
```



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Next Class: Convergence and Multivariate Case

The scalar case ($f: \mathbb{R} \rightarrow \mathbb{R}$) that we have discussed today is simple and easy to visualize. Next class, we will prove that Newton's method converges to roots before generalizing to solving **systems** of linear equations:

$$\left. \begin{array}{l} f_1(x) = f_1(x_1, x_2, \dots, x_n) = 0 \\ x \in \mathbb{R}^n \quad f_2(x_1, \dots, x_n) = 0 \\ f_n(x_1, \dots, x_n) = 0 \end{array} \right\} \begin{array}{l} F = \begin{bmatrix} f_1 \\ \vdots \\ f_n \end{bmatrix} \\ F(x) = 0 \\ x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \end{array}$$

Find $x^* \in \mathbb{R}^n$ s.t. $F(x^*) = 0$.

我们希望求得一个 r ，为函数 f 的root，所以 $f(r) = 0$ ，但是直接求太难，所以我们使用迭代逼近，每一个 x_{k+1} 都比上一个更接近 r 。因为我们假设 r 离 x_k 不远，所以我们在 x_k 处进行泰勒展开，并且忽略高阶误差，得到 $f(x_k) + f'(x_k)(r - x_k) = 0$ ，现在我们假设 r 就是 x_{k+1} ，下一次迭代的结果，然后解这个方程。

但是注意，牛顿法不一定收敛，只有在 r 和 x_k 离得很近的时候收敛。

同时，我们还可以用牛顿法实现除法，只有加减和乘法。

优化问题可以被等价于找local max/min的过程，但是对于一些复杂函数我们压根找不到他们的解析式，可能他们是一个实验得到的图像，我们只知道一些点的数值和其高阶导数，所以我们可以用二阶泰勒展开去模拟他，在 x_0 处（也是initial guess）展开，然后用迭代法找答案。